

Sis! I Shrunk The Features: Lossy Image Compression on Normalization Free Networks

How I Learned to Stop Worrying and Use JXL [2]

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I. INTRODUCTION TO OUR WORK

Ever since humanity has discovered the abuse of shiny rocks [1], we have since strived to deepen our sophistication of the abuse of tools and materials around us. In the spirit of the ancient Persians as paraphrased by Herodicus, "Drunk Then Sober." In the process of refining our idea, the authors have decided to substitute drunkenness with the common more productive practice of academia-induced sleep-deprivation.

As such, we propose the use of using Conventional Image Compression Algorithms (CICA) for storing Conventional Convolutional Neural Network (CCNN) parameters as a set of images. Which is not really that efficient as we cannot do cross layer compression, but as this is SIGBOVIK, maybe a future reader will be interested in bodging very large canvas support.

We use the relatively new (to LLMs) floating point format called fp8, where we anneal the model weights over the course of training the model for a total of 128 epochs. During training the model weights are slowly lerped to fp8 values from fp32. This slowly crushes the values into the limited domain of fp8 compared to saner fp32, or even fp16. Additionally, we re-interpret NaNs as 0 as lossy image compression may introduce any heretical value permitted by the domain. Another modification we implemented is to take the the non-signed part of the bitstring and reverse it. This way, near-zero significands are more likely to be preserved during lossy encoding. Also exponent field is stored in the inferior bits, as no one cares about them.

II. RESULTS

A. Small Model

For the small sized model we use the small version of nfnet provided by timm, this smallest model starts at just 0.178 MiB [3] [4] [5]. JPEG, unsurprisingly, performs the worst of the formats we tested, and doesn't even get close to baseline accuracy by the time it is at similar size to the baseline mode. Second worst is the AVIF format, followed closely by the WebP format. Best performing on the small model is jxl. To note, running JXL in lossy mode with quality 100% is different than running it in lossless mode. lossy mode with 100% quality results in a model that performs slightly worse,

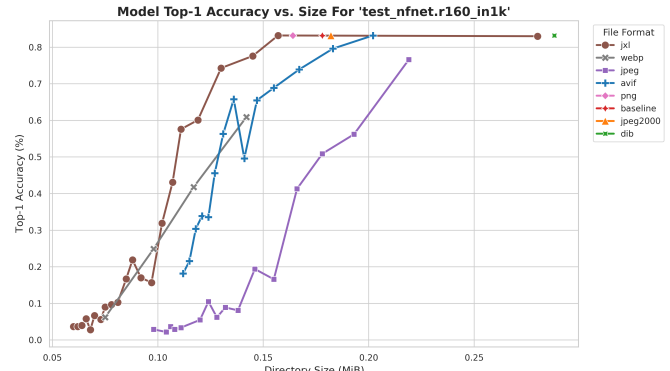


Fig. 1: Small Model, Size vs Top-1

and is significantly larger. Of the lossless formats tested, which means the models have identical accuracy to the baseline, JXL performs the best, then PNG, then the baseline, then lossless AVIF and then, eventually, JXL at 100% quality and finally dib. For a model this small, the main result is that lossless JXL helps a bit, but aggressive image-style compression damages the weights faster than it reduces storage.

B. Medium Model

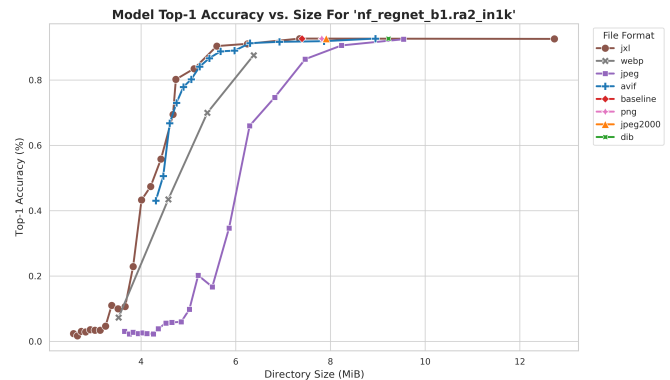


Fig. 2: Medium Model, Size vs Top-1

For the medium sized model we use RegNext pretrained by timm, that has been modified not to use any Normalization

Layers [3] [4] [5]. Lossless JXL again performs the best of the formats with no loss, being the only format to reduce the total size, from 7.4 MiB to 7.347 MiB. all of the other formats increased the size, with both PNG and JPEG 2000 just under 8MiB, and lossless AVIF and dib out around 9 MiB. JXL with quality set to 100% performed markedly worse than everything else, at around 12.7 MiB. Between the lossy formats, JPEG again performed the worst, followed by WebP. AVIF and JXL performed similarly, though JXL regularly slightly outperformed AVIF at the same accuracy level.

C. Large Model

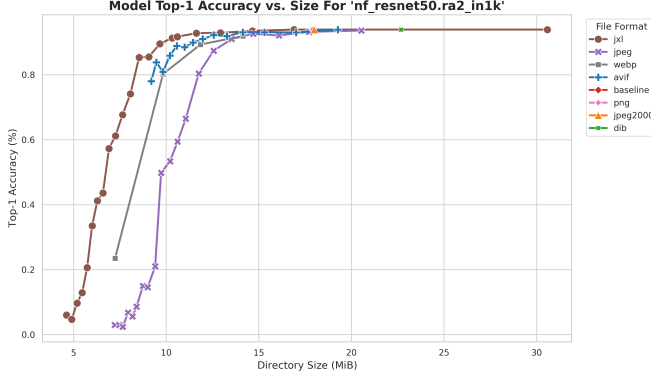


Fig. 3: Large Model, Size vs Top-1

For the large model sizes, we use ResNet50 pretrained by timm, that is also modified without Normalization Layers [3] [4] [5]. The large model is relatively similar to the medium model, in terms of ordering. JPEG once again performs the worse, with a sizeable gap between it and WebP. AVIF is then almost directly in the middle of the gap between WebP and JXL, with JXL once again performing the best. Lossless JXL once again shrinks the model, from 17.9 MiB to 16.9 MiB. PNG only slightly increases the size, at roughly 17.92 MiB, then JPEG2000 at roughly 17.99 MiB. All of the other lossless formats perform significantly worse, with lossless AVIF running at 19.27 MiB, DIB at 22.68 MiB. 100% quality JXL comes in at a whopping 30.575 MiB, around 1.7x the size of the baseline.

D. Lossless

Format	Small	Medium	Large
NONE	0.178	7.4	17.9
png	0.164	7.815	17.916
jxl	0.157	7.347	16.902
jpc	0.182	7.91	17.988
dib	0.288	9.227	22.679

TABLE I: Lossless Sized MiB

Honestly, its works, but doesn't work great. Lossless formats perform the exact same as the baseline, but are larger sized than their lossy cousins, and largely similarly sized to the baseline uncompressed model, As evidenced by Table I, JXL or Jpeg-XL emerges as the most efficient lossless format,

consistently achieving the lowest MiB footprint across small, medium, and large models.

III. SUMMARY

Due to time limitations and the authors' lack of sleep, the full results of the experiments are a Google Sheet in the following folder. Also present are the test models' weights and layers copy pasted from somewhere in our image encoding folder. https://drive.google.com/drive/folders/1of2X_1RlbLoAEB-3PvKTzRJYRd3l9SZq

As such it works, but only for larger models, the smaller model doesn't have enough capacity to survive getting crushed. We found that JXL wins across all sizes, where for a given Top-1 Accuracy value it has the lower model size.

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